

# Arunabha Majumder

**Mechanical / Mechatronics Engineer | Machine Design | FEA | Prototyping & Testing | Control Integration**

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

## PROFESSIONAL SUMMARY

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Mechanical and Mechatronics Engineer with a master's degree (MTech) in Mechatronics, a BTech in Mechanical Engineering and a PhD in progress at Aalborg University, with about three years of hands-on product development in Denmark. Develops and evaluates machine concepts and operating principles, and turns technical requirements into working hardware: 3D models, assemblies and production drawings in SolidWorks, component-level FEA for strength, stiffness and weight, prototype builds, and systematically planned, analysed and documented tests. Understands the interaction between mechanics, sensors, control and software, having specified and implemented the control principles for self-built prototypes. Certified SolidWorks Associate (CSWA); Best Paper Award, IFToMM ISRM 2026.

## CORE SKILLS

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- **Machine Design & 3D CAD:** Machine concepts and operating principles, mechanism and actuator design, 3D models, assemblies, production drawings, SolidWorks (CSWA certified), Fusion 360
- **FEA & Design Optimisation:** Specifying and performing component-level Finite Element Analysis (FEA), analytical stiffness and kinematic modelling, optimising designs for strength, stiffness, weight and assembly (DFA)
- **Prototyping & Testing:** Building prototypes, designing test rigs, planning, conducting, analysing and documenting tests, verification against models
- **Mechanics, Automation & Software:** Control principles and sequences, motor, encoder and force-sensor integration, DAQ, embedded control, pneumatics, PID and model-based control, MATLAB/Simulink, Python
- **Ways of Working:** Translating requirements into designs, systematic documentation, project planning, multidisciplinary collaboration, English (C1)

## PROFESSIONAL EXPERIENCE

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**Research Engineer – Mechanical & Mechatronic Product Development (PhD Project: VIEXO) | 2023 – Present**

*Aalborg University, Aalborg, Denmark*

- Developed and evaluated a new operating principle for a shoulder exoskeleton: a hybrid actuation concept that pairs a parallel spring with a 6 Nm motor, cutting energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Created 3D models, assemblies and production drawings in SolidWorks, and performed component-level FEA to optimise for strength, stiffness and weight before manufacture.
- Built prototypes hands-on from machined and additively manufactured parts and integrated motor, encoder, force sensors, DAQ and embedded control.
- Specified and implemented control principles, including an Assist-as-Needed hybrid controller that reduced user muscle effort by more than 15% in tests.
- Planned, conducted, analysed and documented a 12-participant test campaign across 3 operating modes and 2 load conditions, including test protocol, instrumentation and data pipeline; results support two journal submissions.
- Developed end-effector force estimation for a 5-bar planar parallel robot using variable-stiffness actuators (VSA) and the robot Jacobian, validated against a force sensor to below 10% RMS error.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

**MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022**

*CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Fabricated pneumatic artificial muscle actuators in-house and built the test set-up with valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical model.
- Designed and implemented a neural-network-based gain-scheduled PID controller that reduced position-tracking error to 0.3–0.78% under varying load, outperforming classical PID.

#### **Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019**

*Siddaganga Institute of Technology, Tumakuru, India*

- Led CAD design, fabrication and user testing end to end; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

### **EDUCATION**

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#### **Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present**

*Aalborg University, Department of Materials and Production, Aalborg, Denmark*

#### **MTech (Master's), Mechatronics | 2020 – 2022**

*AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India*

#### **BTech, Mechanical Engineering | 2015 – 2019**

*Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India*

### **CERTIFICATIONS & AWARDS**

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- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019); Best Major Project Award (2019)

### **ENGINEERING PROJECTS**

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- **Go-Kart (2017):** Designed, fabricated and assembled a custom Go-Kart from the ground up.
- **Efficycle (2018):** Designed and built a human-electric hybrid vehicle in a multidisciplinary team for the Efficycle competition.

### **PUBLICATIONS**

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- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, MMS, vol. 213, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

### **TECHNICAL TOOLS**

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- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Control:** MATLAB/Simulink, Python, LabVIEW (basic), Arduino
- **Testing & Instrumentation:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication

### **LANGUAGES**

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English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

### **REFERENCES**

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Available upon request.